NO.4 DIESEL PARTICULATE FILTER INDICATOR LIGHT FLASHES/MESSAGE RELATED DIESEL PARTICULATE FILTER MALFUNCTION INDICATED IN DISPLAY [SKYACTIV-D 1.5]

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4	DIESEL PARTICULATE FILTER INDICATOR LIGHT FLASHES/MESSAGE RELATED DIESEL PARTICULATE FILTER MALFUNCTION INDICATED IN DISPLAY
DESCRIPTION	• Diesel particulate filter indicator light flashes. • Message related diesel particulate filter malfunction indicated in display.
POSSIBLE CAUSE	• PCM DTC is stored. • Exhaust gas pressure sensor No.2 malfunction • Diesel particulate filter clogged • Engine coolant temperature is low (overcooling) • Instrument cluster malfunction • Poor engine oil quality • Poor fuel quality • Abnormal engine combustion **Warning** **• The following troubleshooting flow chart contains the fuel system diagnosis and repair procedures. Read the following warnings before performing the fuel system services:** **— Always keep sparks and flames away from fuel. Fuel can be easily ignited which could cause serious injury or death, and damage to equipment.** **— Fuel line spills and leakage from the pressurized fuel system are dangerous. Fuel can ignite and cause serious injury or death, and damage to property and facilities. Fuel can also irritate skin and eyes. To prevent this, always complete the "Fuel Line Safety Procedure", while referring to the "BEFORE SERVICE PRECAUTION". (See BEFORE SERVICE PRECAUTION [SKYACTIV-D 1.5].)** **— Fuel is highly flammable and dangerous. Fuel line spills and leakage can cause serious injury or death, and damage to equipment. When installing the fuel hose, always refer to the "AFTER SERVICE PRECAUTION" and perform the "Fuel Hose Installation Procedure". (See AFTER SERVICE PRECAUTION [SKYACTIV-D 1.5].)**

Diagnostic Procedure

STEP	INSPECTION	RESULTS	ACTION
1	**VERIFY PCM DTC** • Retrieve PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**DETERMINE IF MALFUNCTION IS DUE TO OVERCOOLING** • Verify the symptom. • Is the cooling system normal?	Yes	Go to the next step.
		No	Perform the symptom troubleshooting "NO.23 COOLING SYSTEM CONCERNS-RUNS COLD". (See NO.23 COOLING SYSTEM CONCERNS-RUNS COLD [SKYACTIV-D 1.5].)
3	**VERIFY PM ACCUMULATION AMOUNT IN DIESEL PARTICULATE FILTER** • Access the following PIDs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) — PM_ACC_DSD — PM_GEN • Start the engine and race it at **3,000 rpm** until the PM_ACC_DSD monitored value stabilizes. • Is the PM_ACC_DSD monitored value twice that of the PM_GEN monitored value or more?	Yes	Go to the next step.
		No	Go to Step 5.
4	**INSPECT EXHAUST GAS PRESSURE SENSOR NO.2** • Inspect the exhaust gas pressure sensor No.2. (See EXHAUST GAS TEMPERATURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is the exhaust gas pressure sensor No.2 normal?	Yes	Go to the next step.
		No	Replace the exhaust gas pressure sensor No.2, then go to Step 10. (See EXHAUST GAS TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

STEP	INSPECTION	RESULTS	ACTION
5	**INSPECT DIESEL PARTICULATE FILTER FOR CLOGGING** • Perform the “COMPULSORY DIESEL PARTICULATE FILTER REGENERATION”. (See COMPULSORY DIESEL PARTICULATE FILTER REGENERATION [SKYACTIV-D 1.5].) • Access the PM_ACC_DSD PID using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the PM_ACC_DSD PID value **approx. 0**?	Yes	Go to Step 7.
		No	Go to the next step.
6	**INSPECT DIESEL PARTICULATE FILTER** • Verify all accessory loads are off. • Access the following PIDs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) — PM_ACC_DSD — PM_GEN • Start the engine and race it for several minutes. • Is the increase in the PM_ACC_DSD and the PM_GEN monitored values the same?	Yes	Go to the next step.
		No	Replace the catalytic converter, then go to Step 10. (See EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
7	**INSPECT INSTRUMENT CLUSTER** • Inspect the instrument cluster. (See INSTRUMENT CLUSTER INSPECTION.) • Is the instrument cluster normal?	Yes	Go to the next step.
		No	Replace the instrument cluster, then go to Step 10. (See INSTRUMENT CLUSTER REMOVAL/INSTALLATION.)
8	**INSPECT FOR MALFUNCTION DUE TO POOR ENGINE OIL** • Replace the engine oil. (See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].) • Does the symptom disappear?	Yes	Advise the customer to have the engine oil replaced periodically. Perform the “ENGINE OIL DATA RESET”. (See ENGINE OIL DATA RESET [SKYACTIV-D 1.5].)
		No	Perform the “ENGINE OIL DATA RESET”, then go to the next step. (See ENGINE OIL DATA RESET [SKYACTIV-D 1.5].)
9	**INSPECT FOR MALFUNCTION DUE TO POOR FUEL** • Replace the fuel. (See FUEL DRAINING PROCEDURE [SKYACTIV-D 1.5].) • Does the symptom disappear?	Yes	Advise the customer as to the change in the fuel used.
		No	Remove the accumulated matter in the cylinder head using the following procedure, then go to the next step. • Carbon remover • Overhauling
10	Verify the test results. • If normal, return to the diagnostic index to service any additional symptoms. (See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].) • If a malfunction remains, inspect the related Service Information and perform the repair or diagnosis. — If the vehicle is repaired, troubleshooting is completed. — If the vehicle is not repaired or additional diagnostic information is not available, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)		